

Tianyang Zhao

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EDUCATION

Pennsylvania State University, University Park, PA

Ph.D. Informatics, GPA: 3.96/4.00

Anticipated: Fall 2027

Pennsylvania State University, University Park, PA

M.S. Computer Science and Engineering, Advisor: Dr. Rui Zhang, GPA: 3.69/4.00

August 2022

Master Thesis: Dialogue State Tracking with Universal Adversarial Triggers

Pennsylvania State University, University Park, PA

B.S. Computer Science, minor in Psychology, GPA: **3.93**/4.00

May 2020

RESEARCH INTERESTS

▪ Natural Language Processing ▪ Trustworthy LLM ▪ Generative AI ▪ Machine Learning

PUBLICATIONS & PROFESSIONAL SERVICES

Publications and Presentations

- Story, P., Ghanavati, S., Hosseini, H., Mitrovic, J., Samples, T., Wagner, I., **Zhao, T.** (2025). Datasets, automation, and analysis methods. In F. Schaub, C. Utz, S. Wilson, & L. Xian (Eds.), *Grand challenges for research on privacy documents (Dagstuhl Seminar 25021)* (pp. 11–16). *Dagstuhl Reports*, 15(1), 1–32. <https://doi.org/10.4230/DagRep.15.1.1>
- **Zhao, T.**, Karimi, Y., Norton, T., Sadeh, N., Wilson, S., Vaniea K. (2025). Questioning Privacy: Contrasting User Questions with Questions Answered by Privacy Policies. *Poster presented at the 12th Mid-Atlantic Student Colloquium on Speech, Language and Learning (MASC-SLL)*.
- Bandyopadhyay, S., **Zhao, T.** (2020). Natural Language Response Generation from SQL with Generalization and Back-translation. In *Proceedings of the First Workshop on Interactive and Executable Semantic Parsing (IntEx-SemPar)* (pp. 42–47).

Peer Reviewer

▪ EMNLP 2023 Industry Track ▪ EMNLP 2022: 3 submissions ▪ NAACL 2022: 1 submission

Conferences and Workshops

Dagstuhl Seminar 25021, Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Wadern, Germany (January 5–10, 2025)

- Invited participant. Identified challenges, research questions, and solution ideas for creating, annotating, and analyzing large-scale research documents.

EMNLP 2020, Virtual Event (November 2020)

- Oral presentation of *Natural language response generation from SQL with generalization and back-translation* at the First Workshop on Interactive and Executable Semantic Parsing (IntEx-SemPar).

RESEARCH EXPERIENCE

Being Stable and Reliable under Twists: Robustness of GraphRAG Systems

Pennsylvania State University

June 2025 – Present

- Evaluated various GraphRAG frameworks' QA performance under different graph modifications.
- Performed case studies and retrieval tracing to analyze and uncover the graph's effect on robustness.

Privacy Questions Elicitation by Generative Large Language Models

Pennsylvania State University

June 2024 – April 2025

- Compared privacy questions from large language models acting as people with different ages and genders.

PrivaSeerQA: A Large-Scale Corpus of Question-Answer Pairs from Privacy Policies

Pennsylvania State University

August 2022 – Present

- Collected questions and responses in privacy policies with an automated approach.
- Developed question-answering systems with various language models, some of which achieving state-of-the-art performance, to show the usability of the corpus and the effect of its large scale.
- Compared the privacy questions addressed in policies and those the general public is most interested in.

Robustness and Augmentation with Universal Adversarial Triggers on Dialogue State Tracking

Pennsylvania State University

May 2021 – June 2022

- Analyzed how Universal Adversarial Triggers influence performance of TripPy on Dialogue State Tracking.
- Investigated if augmenting data with these triggers can improve TripPy's performance.

Efficient Sub-Pixel Convolutional Neural Network for Spectrum Image Processing

Pennsylvania State University

December 2020 – May 2021

- Reviewed related literature to find approaches for pattern recognition.
- Implemented these ideas, ran analysis experiments, and evaluated their accuracy as well as efficiency.

Adversarial Machine Learning with SQL-to-Text Translation

Independent Research Project

May 2020 – August 2020

- Improved the robustness of a model by training with extra adversarial data.
- Identified common types of noise and generated noisy datasets corresponding to each type.
- Designed various character perturbations and compared effects on original and noisy datasets.

SELECTED PROJECTS

New York City Pick-up and Drop-off Demands Prediction Tool

- Built a web application using HTML for front-end web pages and Flask for the back-end interactions.
- Carried out data analysis to clean the data, discover useful features, and generate visualizations.
- Compared various approaches: random forest, nearest neighbors, ARIMA, and linear regression with Lasso.

CanvasPath: A Course Management System with Database Support

- Conducted requirement analysis to find out the expected functionalities and the best suited development tools.
- Designed a database to capture the requirements, expressed by an entity relationship diagram.
- Implemented the system as a web application with different interfaces for various types of users.

SKILLS

- Selected Deep Learning Python Libraries: PyTorch, Keras, TensorFlow, scikit-learn, Transformers
- Selected Data Processing Python Libraries: NLTK, NumPy, Pandas, matplotlib, SciPy
- Large Language Model Frameworks: Ollama, LitGPT, LangChain
- Programming Languages: Python, Matlab, C, Java, C++
- Web Development: SQL, HTML, JavaScript, CSS, Flask
- Miscellaneous: LaTeX, Git, Jupyter Notebook

TEACHING EXPERIENCE

Teaching Assistant, Pennsylvania State University, College of IST

Statistical Analysis for Information Sciences

August 2025 – December 2025

Intermediate & Object-Oriented Application Development

August 2022 – December 2022

Teaching Assistant, Pennsylvania State University, Department of CSE

Data Structures and Algorithms

Discrete Math for Computer Science

Programming and Computation I: Fundamentals

August 2021 – May 2022

January 2021 – May 2021

August 2020 – December 2020